THE SIMON85 PROJECT

CPM85

Pocket Guide

HP-85 BASIC control of native CP/M

LOADBIN CPM85, v0.06d
Requires matching EBTKS and ESP32 B2d firmware

An independent HP-85 extension reference

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Getting Started

CPM85 adds BASIC statements for using a parallel Z80 CP/M 2.2 environment, managing mounted disk images, and transferring files to/from the Teensy SD card. **EBTKS** Use firmware with the native Z80 CP/M bridge and HP drive-control mailbox; place CP/M disk images under `/cpm/disks` and transfer files under `/cpm/transfers` on the Teensy SD card. Load the CPM85 binary with `LOAD BIN "CPM85"` (adjust the host file name to your installation). If PRINTER85 has captured keyboard keys, run `PRTKEYS 0` first. **First test** With CP/M at its `A>` prompt, enter `CPVER`, `CPSTATUS` and `CPM`. Press K8 to return to HP BASIC; CP/M continues running.

Console & Status

CPM

`CPM`

Connects the HP-85 keyboard to CP/M. Shows `CONNECTED TO CP/M - K8 TO EXIT`. K8 detaches keyboard without stopping CP/M.

`CPM`

CPOFF

`CPOFF`

Releases CPM85 keyboard capture (when HP BASIC control is available).

`CPOFF`

CPVER

`CPVER`

Reports the installed CPM85 binary version.

```
CPVER
```

CPCOUNT

`CPCOUNT`

Numeric function returning number of captured keys modulo 256.

```
PRINT CPCOUNT
```

CPLAST

`CPLAST`

Numeric function returning the last raw HP keyboard code, including K8.

```
PRINT CPLAST
```

CPSTATUS

`CPSTATUS`

Lists active and pending CP/M drive-image assignments.

```
CPSTATUS
```

Disk Images

CPDIMGS

`CPDIMGS`

Lists available disk image filenames in the SD-card `/cpm/disks` directory.

```
CPDIMGS
```

CPDMK

```
CPDMK "image.dsk"
```

Creates a new empty 8 MiB CP/M hard-disk image on SD, without replacing an existing one.

```
CPDMK "WORK.DSK"
```

CPDMNT

```
CPDMNT "drive:","image.dsk"
```

Stages a disk-image mount for the selected CP/M drive. Apply with CPRESET while CP/M is idle.

```
CPDMNT "F:","WORK.DSK"
```

CPDUMNT

```
CPDUMNT "drive:"
```

Stages unmount of a CP/M drive. Requires CPRESET to apply.

```
CPDUMNT "F:"
```

CPRESET

```
CPRESET
```

Resets the emulated Z80/CP/M system and applies pending disk assignments; HP-85 itself continues running. Save CP/M work first.

```
CPRESET
```

CPREBOOT

```
CPREBOOT
```

Reboots the ESP32 and its services. HP-85 remains powered on; CP/M and Wi-Fi sessions are interrupted.

```
CPREBOOT
```

CPDCAT

```
CPDCAT "drive:"
```

Displays the selected CP/M drive directory using native read-only BDOS access. Run at an idle CP/M command prompt.

```
CPDCAT "A:"
```

File Transfer

Transfers are blocking HP BASIC statements. They complete silently on success. CP/M must be idle at its command prompt; no active application may be using BDOS. Existing destination files are never overwritten.

CPEXPORT

```
CPEXPORT "drive:file"[, "sd-name"]
```

Copies one CP/M 8.3 file to the Teensy SD-card `/cpm/transfers` directory. Default SD filename is the drive prefix plus underscore (for example `A_PIP.COM`); optional second string gives a different relative SD name.

```
CPEXPORT "A:PIP.COM","PIPB2D.COM"
```

CPIMPORT

```
CPIMPORT "sd-name","drive:file"
```

Copies one literal SD transfer file into a NEW CP/M 8.3 destination. The first operand is an SD filename (no CP/M drive colon).

```
CPIMPORT "PIPB2D.COM","F:PIP.COM"
```

CPXCODE

`CPXCODE`

Numeric function giving the last transfer result: 0 success; 1 CP/M not idle; 2 file operation failed; 3 transport/transaction uncertain; 4 transfer entered BDOS (normally replaced by terminal result).

```
PRINT CPXCODE
```

CPXSTAT

`CPXSTAT`

Displays diagnostic status of the outstanding/most recent transfer, especially after BASIC STOP or a communication failure.

```
CPXSTAT
```

CPABORT

`CPABORT`

Requests cancellation of a pending transfer before guest BDOS starts. Does NOT interrupt a running BDOS file operation.

```
CPABORT
```

Errors & Safety

ERROR 74 (STALL) with `CPXCODE=1` means CP/M is not idle. Confirm the command prompt, then retry. The cancelled transfer does not create a file later.

ERROR 67 (FILE NAME) with `CPXCODE=2` reports a file/storage failure, including attempted overwrite of an existing destination.

CPXCODE=3 means transport/transaction status is uncertain. Inspect `CPXSTAT`; do not restart or reset an unresolved transfer blindly.

HP BASIC programs can trap native errors using `ON ERROR GOTO` or `ON ERROR GOSUB` and inspect `ERRN`. Check the HP-85 Pocket Guide for normal BASIC handler behavior.

The three CPM85 transfer operations rely on the matching B2d ESP32 and Teensy protocols. Do not mix firmware versions when updating.